

Chapter 7

Electromagnetic Waves



Lesson outlines:

7-1 Electromagnetic Waves

7-2 Electromagnetic Waves, and Their Speed

**7-3 Light as an Electromagnetic Wave and the
Electromagnetic Spectrum**

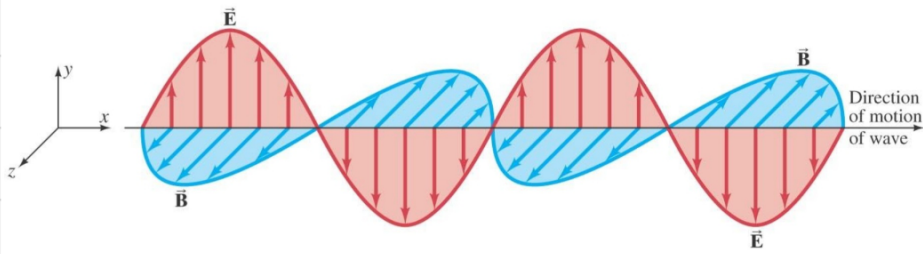
7-1 Electromagnetic Waves

Since a changing electric field produces a magnetic field, and a changing magnetic field produces an electric field, once sinusoidal fields are created they can propagate on their own.

These propagating fields are called **electromagnetic waves**.

Electromagnetic waves

- electric waves, $\vec{E}$ is perpendicular to magnetic waves, $\vec{B}$



- $\frac{E}{B} = v$
- $v = c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = 3.0 \times 10^8 \text{ m s}^{-1}$
- $v = f\lambda$, $k = \frac{2\pi}{\lambda}$, $\omega = 2\pi f$, $v = \frac{\omega}{k}$
- $\vec{E} = E_y = E_0 \sin(kx - \omega t)$
- $\vec{B} = B_z = B_0 \sin(kx - \omega t)$

7-1 Electromagnetic Waves

The electric and magnetic waves are perpendicular to each other, and to the direction of propagation.

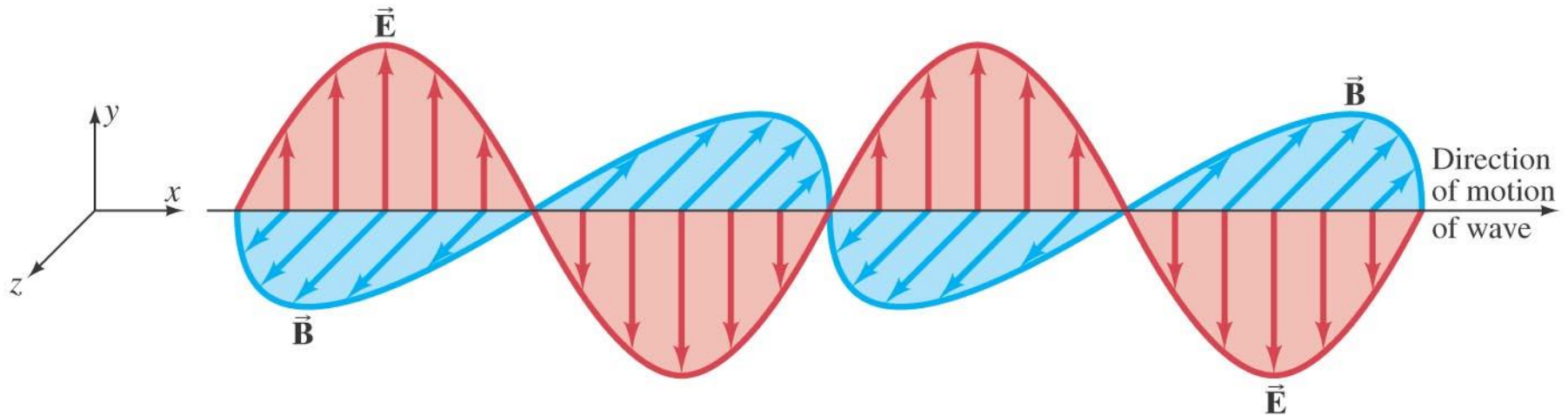


Figure 31-9. Electric and magnetic field strengths in an electromagnetic wave. E and B are at right angles to each other. The entire pattern moves in a direction perpendicular to both E and B .

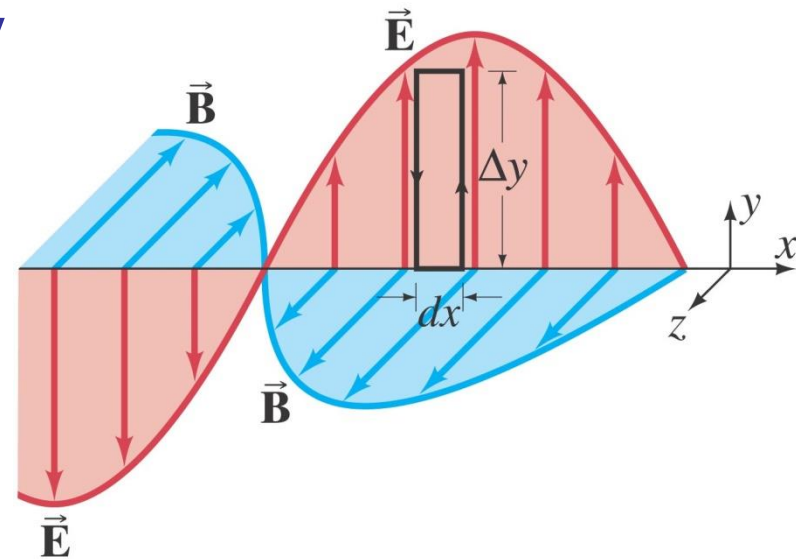
7-2 Electromagnetic Waves, and Their Speed

This figure shows an electromagnetic wave of wavelength λ and frequency f . **The electric and magnetic fields are given by**

$$E = E_y = E_0 \sin(kx - \omega t)$$
$$B = B_z = B_0 \sin(kx - \omega t)$$

where

$$k = \frac{2\pi}{\lambda}, \quad \omega = 2\pi f, \quad \text{and} \quad f\lambda = \frac{\omega}{k} = v.$$



7-2 Electromagnetic Waves, and Their Speed

$$\frac{E}{B} = v.$$

Here, v is the velocity of the wave.
Substituting,

$$v = c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}.$$

7-2 Electromagnetic Waves, and Their Speed, Derived from Maxwell's Equations

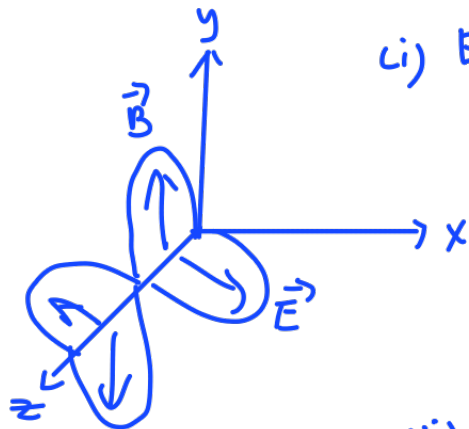
The magnitude of this speed is 3.0×10^8 m/s – precisely equal to the measured speed of light.

7-2 Electromagnetic Waves, and Their Speed

Example 1: Determining $\vec{E}$ and $\vec{B}$ in EM waves.

$$f = 60 \text{ Hz}$$

Assume a 60-Hz EM wave is a sinusoidal wave propagating in the z direction with $\vec{E}$ pointing in the x direction, and $E_0 = 2.0 \text{ V/m}$. Write vector expressions for $\vec{E}$ and $\vec{B}$ as functions of position and time.



$$\begin{aligned} \text{ci) } E_x &= E_0 \sin(kz - \omega t) \\ &= 2.0 \sin(1.25 \times 10^{-6} z - 120\pi t) \end{aligned}$$

$$\begin{aligned} \omega &= 2\pi f & v &= f\lambda \\ &= 2\pi(60) & 3 \times 10^8 &= 60(\lambda) \\ &= 120\pi \text{ rad/s} & \lambda &= 5 \times 10^6 \end{aligned}$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{5 \times 10^6} = 1.25 \times 10^{-6}$$

$$\text{cii) } v = \frac{E_0}{B_0}$$

$$3 \times 10^8 = \frac{2}{B_0}$$

$$B_0 = 6.67 \times 10^{-9} \text{ T}$$

$$\begin{aligned} B_y &= B_0 \sin(kz - \omega t) \\ &= 6.67 \times 10^{-9} \sin(1.25 \times 10^{-6} z - 120\pi t) \end{aligned}$$

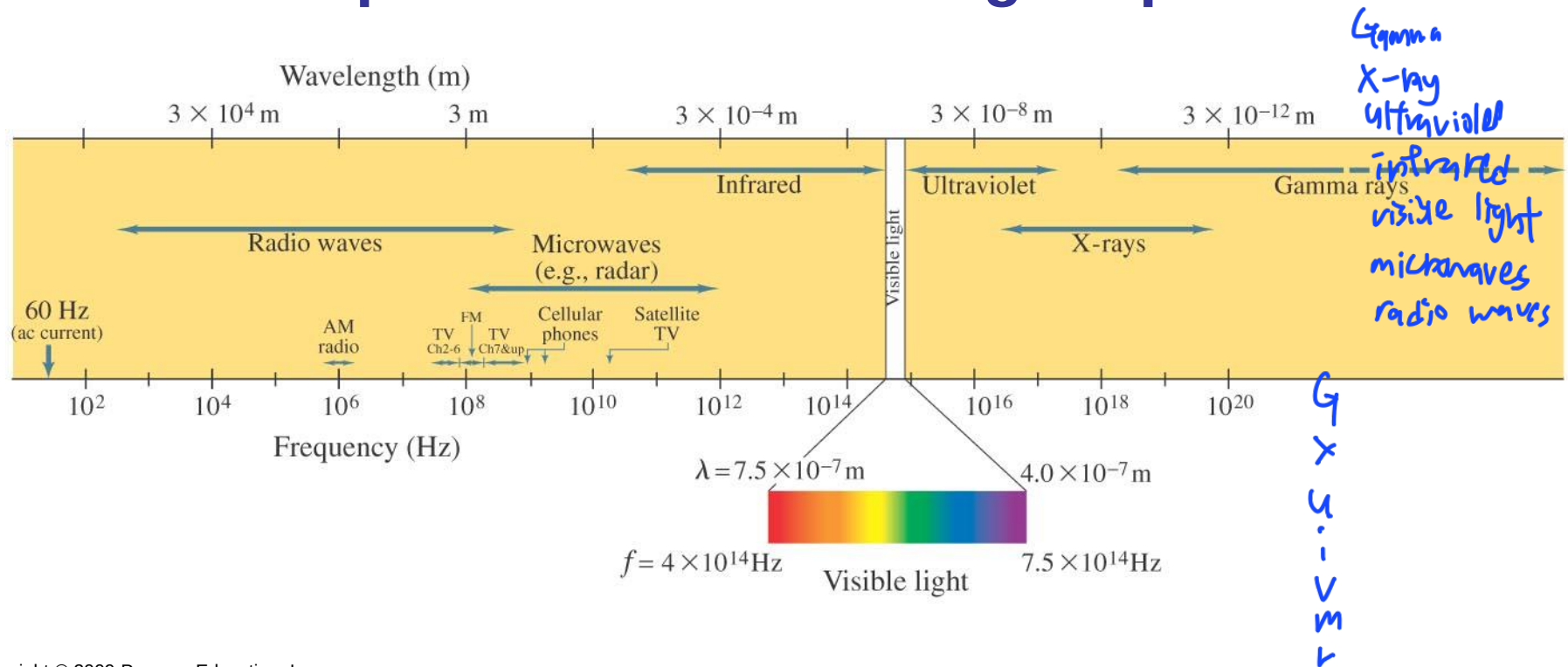
7-3 Light as an Electromagnetic Wave and the Electromagnetic Spectrum

The frequency of an electromagnetic wave is related to its wavelength and to the speed of light:

$$c = \lambda f.$$

7-3 Light as an Electromagnetic Wave and the Electromagnetic Spectrum

Electromagnetic waves can have any wavelength; we have given different names to different parts of the wavelength spectrum.



7-3 Light as an Electromagnetic Wave and the Electromagnetic Spectrum

Example 2: Wavelengths of EM waves.

Calculate the wavelength

(a) of a 60-Hz EM wave,

(b) of a 93.3-MHz FM radio wave, and

(c) of a beam of visible red light from a laser at frequency 4.74×10^{14} Hz.

a. 5.0×10^6 m

b. 3.22 m

c. 6.33×10^{-7} m

$$(a) \quad v = f\lambda$$

$$3 \times 10^8 = 60\lambda$$

$$\lambda = 5.0 \times 10^6 \text{ m}$$

$$(b) \quad v = f\lambda$$

$$3 \times 10^8 = 93.3 \times 10^6 (\lambda)$$

$$\lambda = 3.22 \text{ m}$$

$$(c) \quad v = f\lambda$$

$$3 \times 10^8 = 4.74 \times 10^{14} (\lambda)$$

$$\lambda = 6.33 \times 10^{-7} \text{ m}$$

7-3 Light as an Electromagnetic Wave and the Electromagnetic Spectrum

Example 3: Cell phone antenna.

The antenna of a cell phone is often $\frac{1}{4}$ wavelength long. A particular cell phone has an 8.5-cm-long straight rod for its antenna. Estimate the operating frequency of this phone.

$$8.5 \times 10^{-2} = 34 \text{ cm} \\ = 0.34 \text{ m}$$

$$\lambda = 0.34 \text{ m}$$

$$v = f\lambda \\ 3 \times 10^8 = f(0.34) \\ f = 8.82 \times 10^8 \text{ Hz}$$